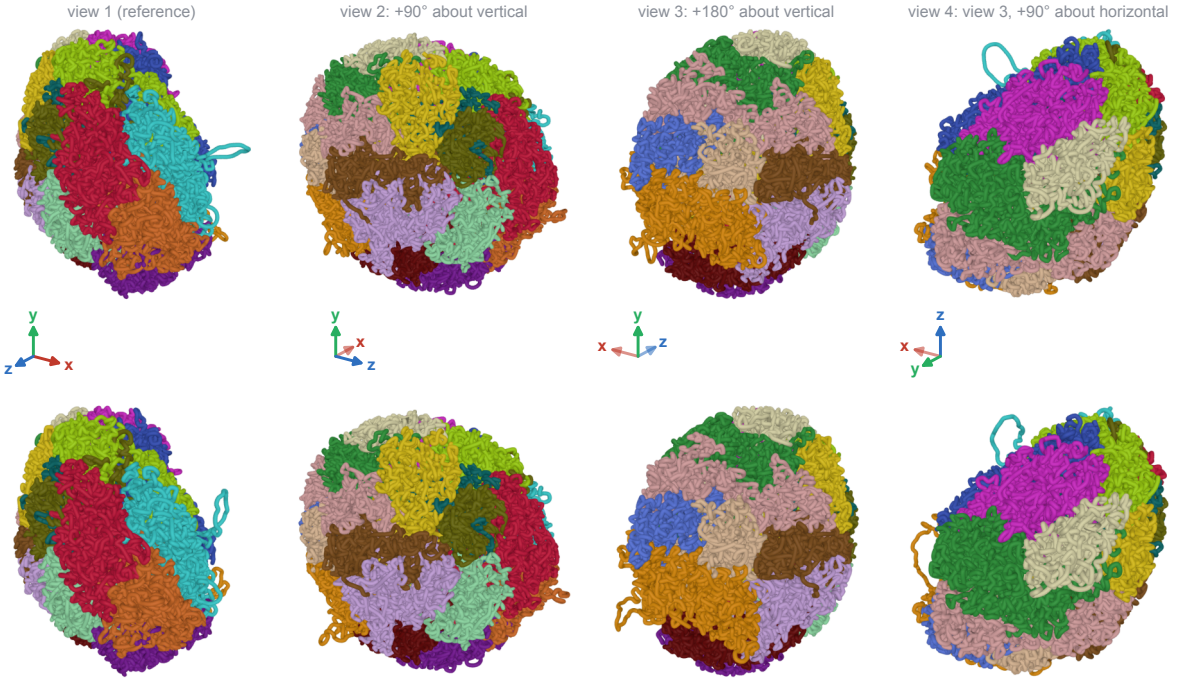
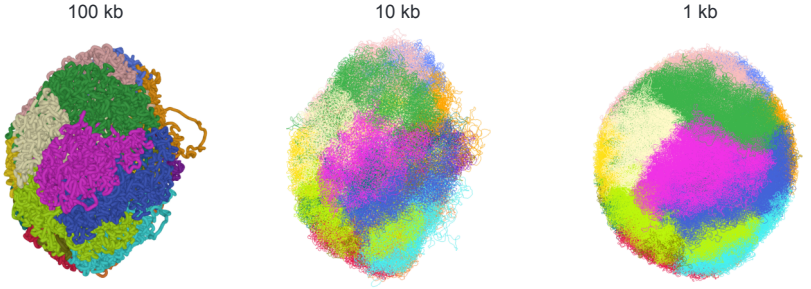
Stevens 2017 Cell 1, 100 kb; coloured by chromosome; the same camera in both rows; axes glyphs: the structure's x / y / z after each rotation (approximate)



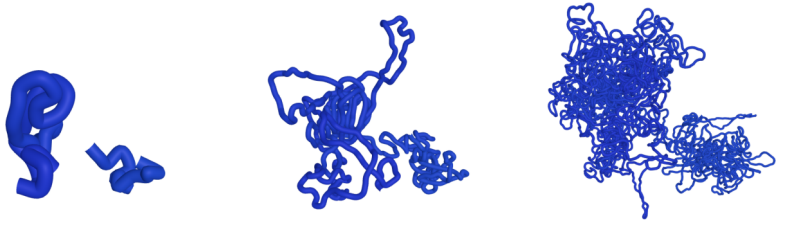
a **Whole genome**

Stevens 2017 Cell 1, model 1 of 10, coloured by chromosome



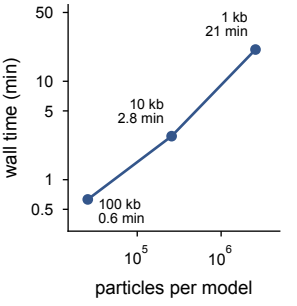
b **chr1:30–35 Mb**

same model, zoomed

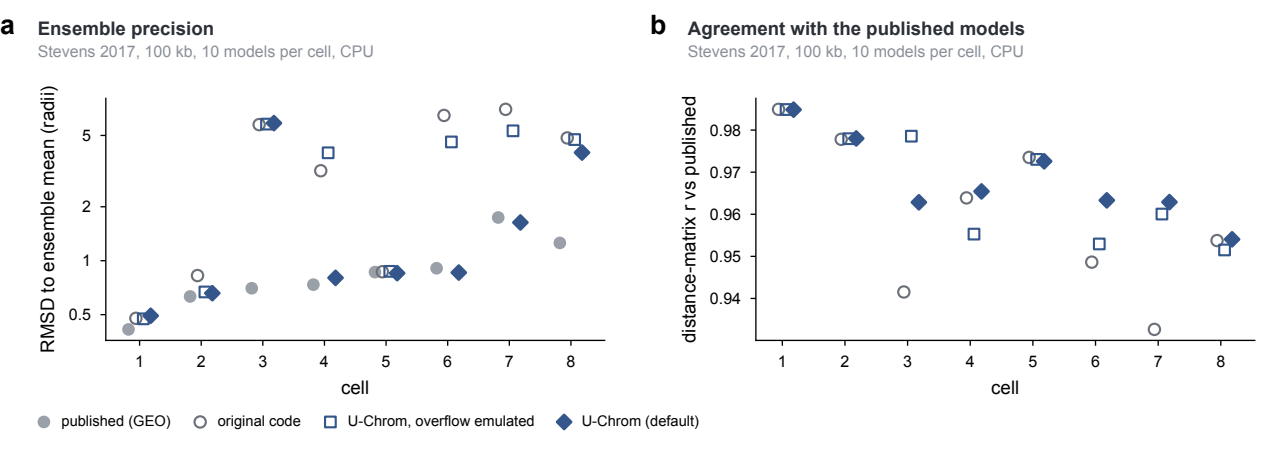


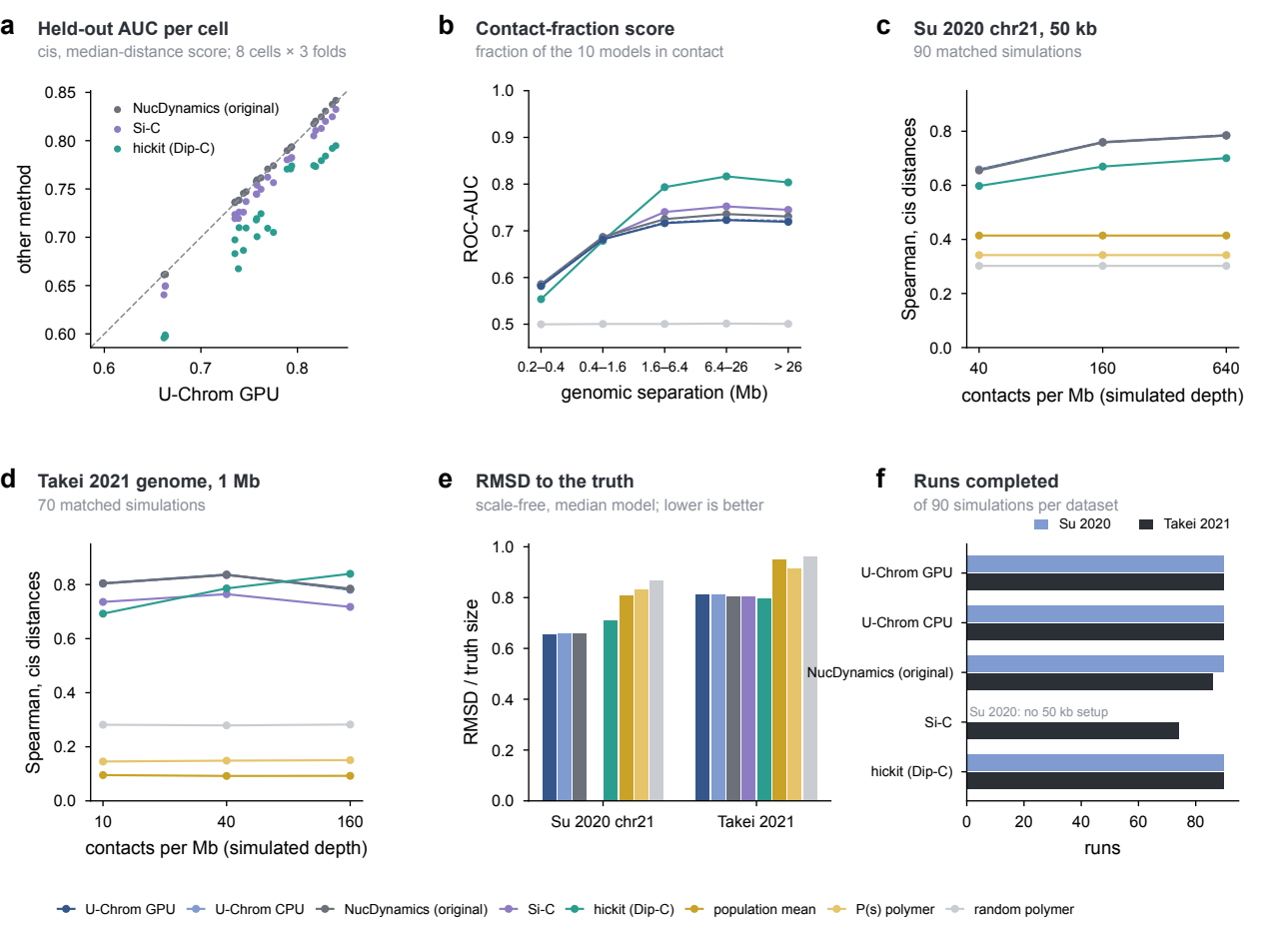
c **Cost on one GPU**

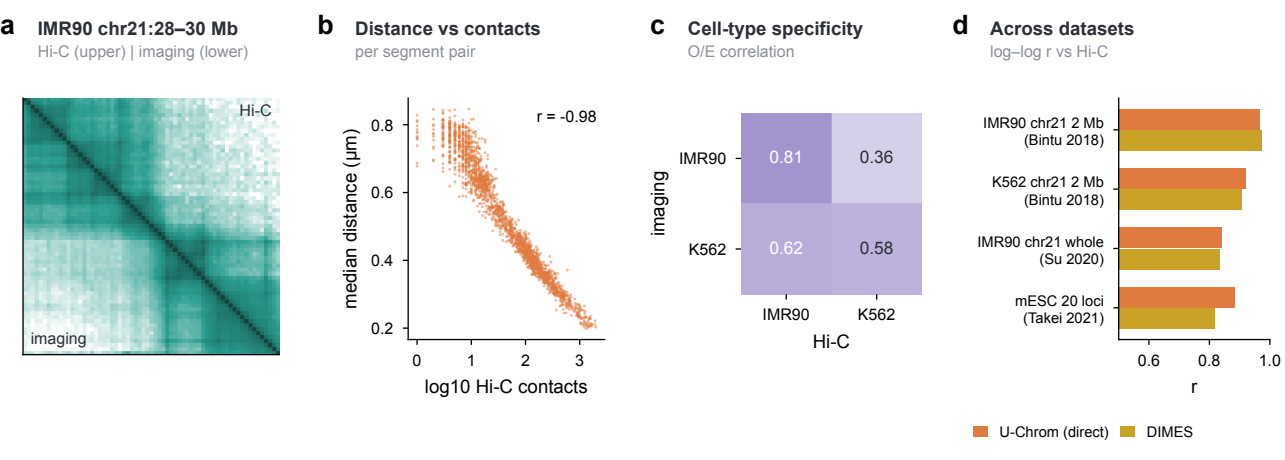
10 models, full schedule

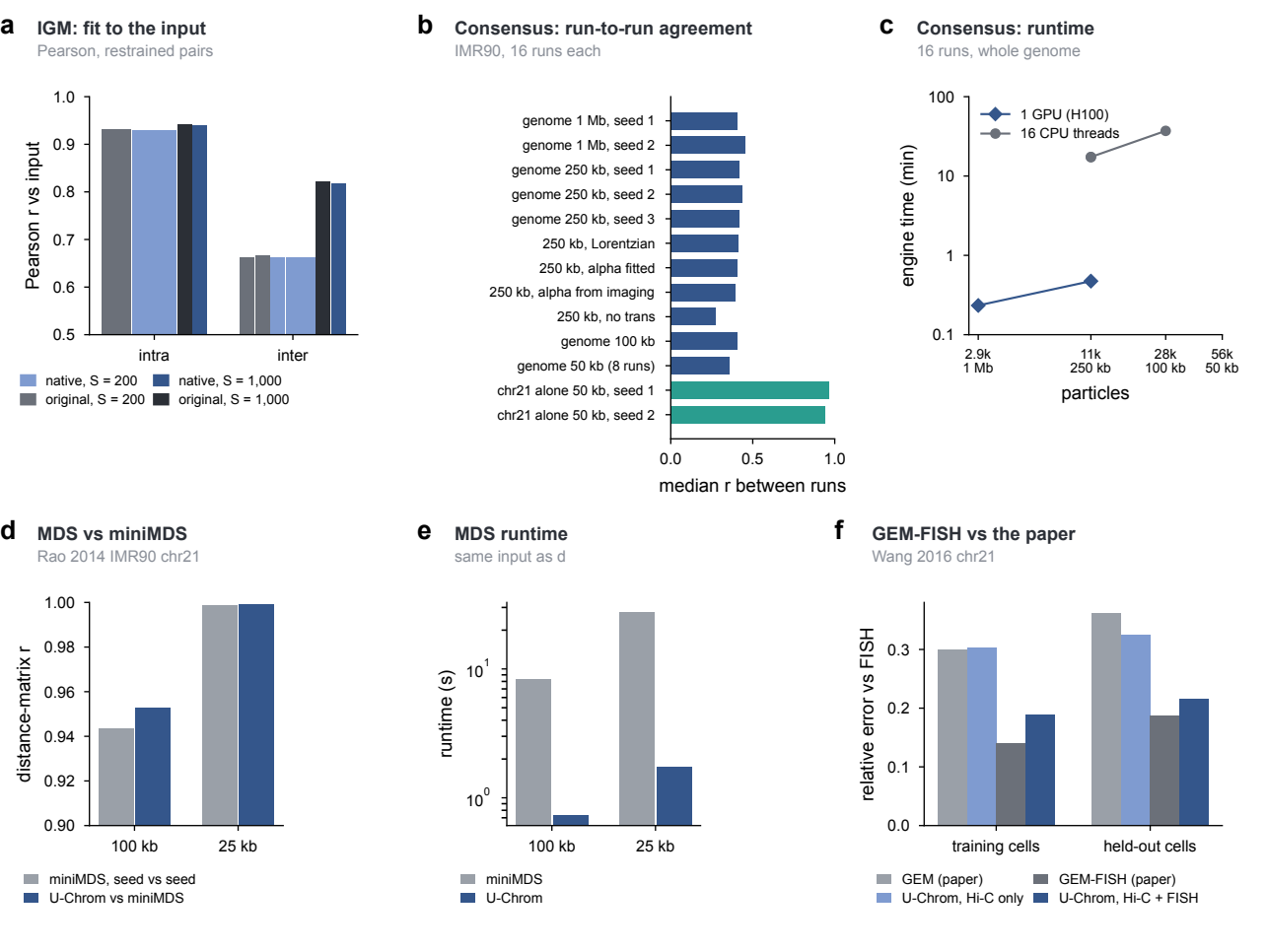


NVIDIA H100, one GPU
100 kb: 25,724 particles, 0.6 min
10 kb: 256,904 particles, 2.8 min
1 kb: 2,568,721 particles, 21.0 min


a
Ensemble precision
Stevens 2017, 100 kb, 10 models per cell, CPU
RMSD to ensemble mean (radii)
0.5
1
2
5
1
2
3
4
5
6
7
8
cell
b
Agreement with the published models
Stevens 2017, 100 kb, 10 models per cell, CPU
distance-matrix r vs published
0.94
0.95
0.96
0.97
0.98
1
2
3
4
5
6
7
8
cell
published (GEO)
original code
U-Chrom, overflow emulated
U-Chrom (default)


a
Held-out AUC per cell
cis, median-distance score; 8 cells × 3 folds
NucDynamics (original)
Si-C
hickit (Dip-C)
other method
U-Chrom GPU
b
Contact-fraction score
fraction of the 10 models in contact
ROC-AUC
0.2–0.4
0.4–1.6
1.6–6.4
6.4–26
> 26
genomic separation (Mb)
c
Su 2020 chr21, 50 kb
90 matched simulations
Spearman, cis distances
contacts per Mb (simulated depth)
d
Takei 2021 genome, 1 Mb
70 matched simulations
Spearman, cis distances
contacts per Mb (simulated depth)
e
RMSD to the truth
scale-free, median model; lower is better
RMSD / truth size
Su 2020 chr21
Takei 2021
f
Runs completed
of 90 simulations per dataset
Su 2020
Takei 2021
U-Chrom GPU
U-Chrom CPU
NucDynamics (original)
Si-C
Su 2020: no 50 kb setup
hickit (Dip-C)
runs
U-Chrom GPU
U-Chrom CPU
NucDynamics (original)
Si-C
hickit (Dip-C)
population mean
P(s) polymer
random polymer


a
IMR90 chr21:28–30 Mb
Hi-C (upper) | imaging (lower)
Hi-C
imaging
b
Distance vs contacts
per segment pair
r = -0.98
median distance (µm)
0.8
0.6
0.4
0.2
0
1
2
3
log10 Hi-C contacts
c
Cell-type specificity
O/E correlation
imaging
IMR90
K562
0.81
0.36
0.62
0.58
IMR90
K562
Hi-C
d
Across datasets
log–log r vs Hi-C
IMR90 chr21 2 Mb
(Bintu 2018)
K562 chr21 2 Mb
(Bintu 2018)
IMR90 chr21 whole
(Su 2020)
mESC 20 loci
(Takei 2021)
0.6
0.8
1.0
r
U-Chrom (direct)
DIMES


a
IGM: fit to the input
Pearson, restrained pairs
Pearson r vs input
1.0
0.9
0.8
0.7
0.6
0.5
intra
inter
native, S = 200
native, S = 1,000
original, S = 200
original, S = 1,000
b
Consensus: run-to-run agreement
IMR90, 16 runs each
genome 1 Mb, seed 1
genome 1 Mb, seed 2
genome 250 kb, seed 1
genome 250 kb, seed 2
genome 250 kb, seed 3
250 kb, Lorentzian
250 kb, alpha fitted
250 kb, alpha from imaging
250 kb, no trans
genome 100 kb
genome 50 kb (8 runs)
chr21 alone 50 kb, seed 1
chr21 alone 50 kb, seed 2
0.0
0.5
1.0
median r between runs
c
Consensus: runtime
16 runs, whole genome
1 GPU (H100)
16 CPU threads
engine time (min)
100
10
1
0.1
2.9k
1 Mb
11k
250 kb
28k
100 kb
56k
50 kb
particles
d
MDS vs miniMDS
Rao 2014 IMR90 chr21
distance-matrix r
1.00
0.98
0.96
0.94
0.92
0.90
100 kb
25 kb
miniMDS, seed vs seed
U-Chrom vs miniMDS
e
MDS runtime
same input as d
runtime (s)
10^1
10^0
100 kb
25 kb
miniMDS
U-Chrom
f
GEM-FISH vs the paper
Wang 2016 chr21
relative error vs FISH
0.3
0.2
0.1
0.0
training cells
held-out cells
GEM (paper)
U-Chrom, Hi-C only
GEM-FISH (paper)
U-Chrom, Hi-C + FISH